

VOLUMETRIC LUNG TUMOR SEGMENTATION

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A project report submitted to

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SCHOOL OF ELECTRONICS ENGINEERING

in partial fulfillment of the requirements for the course of

ECE3009 – Neural Networks and Fuzzy Control

in

**B.Tech. ELECTRONICS AND COMMUNICATION
ENGINEERING**



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NOVEMBER 2023

ABSTRACT

Lung cancer is a leading cause of global mortality, with millions succumbing to its effects annually. Timely and accurate segmentation of lung tumors is critical for early diagnosis and effective treatment planning. This project hopes to segment lung tumors using a volumetric dataset acquired through 3D lung CT scans represented in slice format.

The significance of this project lies in its potential to revolutionize lung cancer diagnostics, enabling precise identification of tumor boundaries and affected regions. Global statistics underscore the urgency, with lung cancer ranking high among fatal malignancies. The dataset, comprising 3D Lung CT Scans from diverse patients, was preprocessed and augmented in order to produce more accurate results. Techniques such as rotation, scaling, and normalization were applied to enhance data quality and facilitate robust model training. The chosen deep learning model, UNet, is well-suited for semantic segmentation tasks. Leveraging its architecture, our approach aims to accurately depict lung tumor boundaries, providing doctors with valuable insights into affected areas. The final goal is to improve diagnostic accuracy and enable targeted interventions, ultimately contributing to enhanced patient outcomes in the fight against lung cancer.

CHAPTER 1

INTRODUCTION

Lung cancer remains a global health challenge, claiming a staggering number of lives annually. According to the World Health Organization (WHO), lung cancer is responsible for approximately 2.2 million deaths each year, making it the leading cause of cancer-related mortality worldwide. This alarming statistic underscores the urgent need for advancements in early detection and precise treatment strategies. Lung tumor Segmentation, the process of identifying and delineating cancerous regions within the lungs, emerges as a crucial aspect in addressing this global health concern.

This project centers on the significance of Lung tumor Segmentation, aiming to enhance diagnostic precision and treatment planning for lung cancer patients. With the mortality rates associated with lung tumors reaching unprecedented levels, efficient and accurate segmentation is paramount. By leveraging cutting-edge deep learning techniques, specifically the UNet model, this project seeks to revolutionize the way we identify and understand lung tumors, ultimately contributing to improved patient outcomes. The dataset forms the cornerstone of our research, comprising 3D Lung CT Scans obtained from a diverse set of patients. These scans, represented in the form of slices, offer a comprehensive view of the lungs in three dimensions. This multi-patient dataset provides a rich source of information crucial for training our deep learning model and ensuring its generalization across various cases.

To prepare the dataset for effective model training, rigorous preprocessing techniques were applied. Rotation, scaling, and normalization processes were implemented to enhance the quality and consistency of the data. Rotation ensures that the model learns from variations in orientation, scaling aids in standardizing the size of the images, and normalization is crucial for maintaining uniformity in pixel values, optimizing the learning process for the deep neural network.

The chosen deep learning model, UNet, is a convolutional neural network (CNN) renowned for its effectiveness in semantic segmentation tasks. UNet excels in capturing intricate details and spatial relationships within images, making it well-suited for our objective of accurately delineating lung tumors. By employing UNet, we aim to facilitate precise identification of affected areas and guiding personalized treatment strategies. In essence, this project represents a pivotal step towards leveraging advanced technology to combat the global health crisis posed by lung cancer.

CHAPTER 2

LITERATURE REVIEW

Author	Title	Year of Publication	Methodology	Inference of the paper
Clement Zotti, Zhiming Luo, Alain Lalande, Pierre-Marc Jodin	Convolutional Neural Network with Shape Prior Applied to Cardiac MRI segmentation	2019	The CNN architecture in this paper consists of five convolutional layers and three fully connected layers. The convolution layers are used to extract features and classification is performed by the fully connected layers. The loss function used was cross-entropy loss.	The proposed model achieves high accuracy for the cardiac MRI scans. It showed high efficiency due to use of a shape prior, which increased the accuracy of segmentation. This model is suitable for real-time applications.
Phi Vu Tran	A Fully Convolutional Neural Network for Cardiac Segmentation in Short-Axis MRI	2016	The Fully Convolutional Network is trained with cross entropy loss function. The model achieves high accuracy on a wide range of image segmentation benchmarks.	The paper implements a Fully Convolutional Network for semantic segmentation. The network learns to segment the image directly from the input data without any specific features. It is computationally efficient and can be used to perform segmentation in real-time.
S. Pereira, A. Pinto, V. Alves, C. A. Silva	Brain tumor Segmentation Using Convolutional Neural Networks in MRI Images	2016	The CNN in this paper gives the option for deep architectures and prevents overfitting due to use of small 3x3 kernels. The network was trained with images whose intensity was normalized for better results.	Using small kernels helped the authors to build deeper architectures with high efficiency and lower computational intensity.

Muhammad Irfan Sharif, Jian Ping Li, Muhammad Attique Khan, Muhammad Asim Saleem	Active deep neural network features selection for segmentation and recognition of brain tumors MRI images	2020	The images in the dataset are preprocessed using Pixel Increase Along Limit (PIaL) technique, which increases the visibility of the tumor. Segmentation of tumors is performed using saliency-based deep learning methods. This method uses a combination of deep learning and texture features which segment the tumor.	The paper uses Pixel Increase along Limit technique to increase contrast and enhance the visibility of the tumor. Segmentation is performed using saliency-based deep learning and a particle swarm optimization algorithm is used to select the most relevant features for classification.
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CHAPTER 3

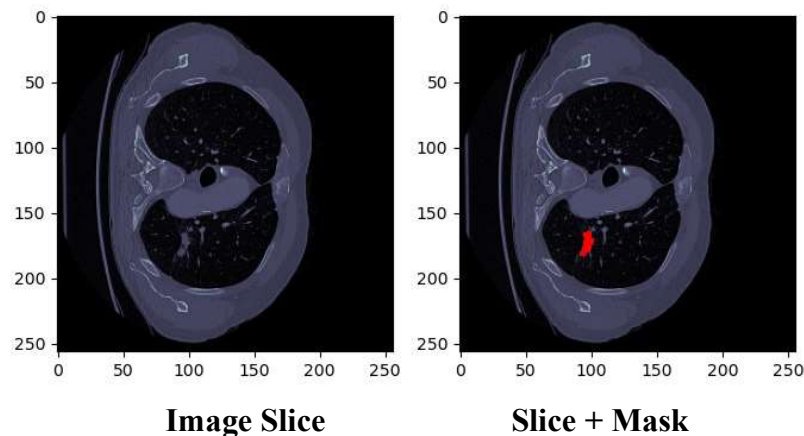
METHODS AND MATERIALS

3.1 MATERIALS

To implement Lung tumor Segmentation, we first have to understand how to operate the 3D dataset of the lungs. Before passing it through the model, we have to preprocess the dataset.

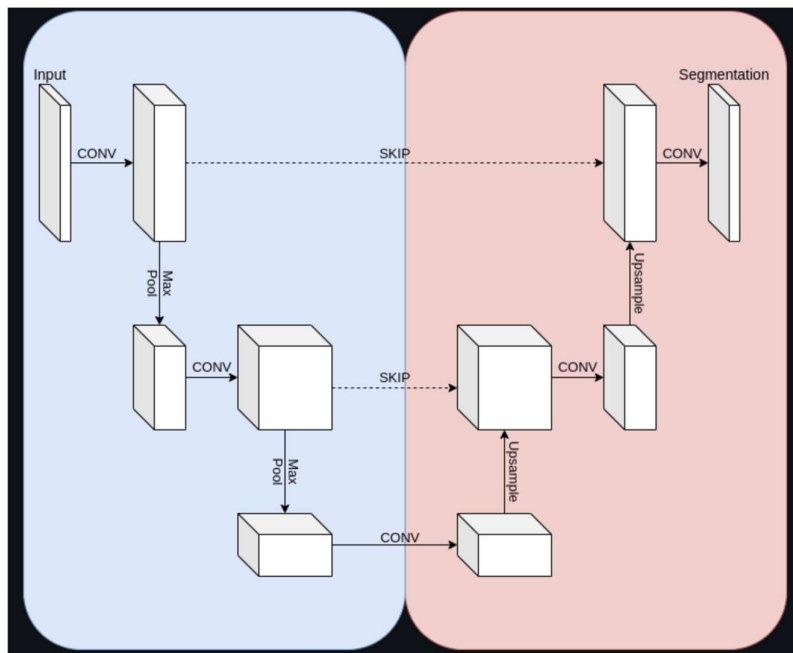
- Pre-processing:
 - ❖ The fixed range of CT scans is -1000 to 3071. Therefore, we don't need to compute the mean and standard deviation for this work because we can normalize by dividing by 3071.
 - ❖ Portions of the lower abdomen can be removed in order to concentrate on lung cancers, which will simplify the image and facilitate network learning. For instance, we may omit the first 30 slices (last axis) and go straight from the lower belly to the neck.
 - ❖ Because reading a single slice is significantly faster than loading the whole NIfTI file, we have saved the preprocessed data as 2D files in order to lower the computational cost of tackling this work on a slice level (2D) rather than a subject level (3D).
 - ❖ Last, we resized the files to (256,256), which is a general image ratio for UNet input layer.
- Dataset Creation:

After implementing the preprocessing methodology step by step, we have returned the image slice and the mask for the tumor observed in the image.



- Model:

The processes dataset is passed down to the standard UNet model. An Encoder-Decoder architecture with extra skip-connections on various levels is the foundation of a UNET: The encoder uses downconvolutional layers to minimize the feature map sizes. By upsampling, the decoder recreates a mask of the input form over many layers. Skip-connections also enable direct information flow on all UNET intermediate levels from the encoder to the decoder. This makes it possible to build masks of excellent quality and makes training easier.



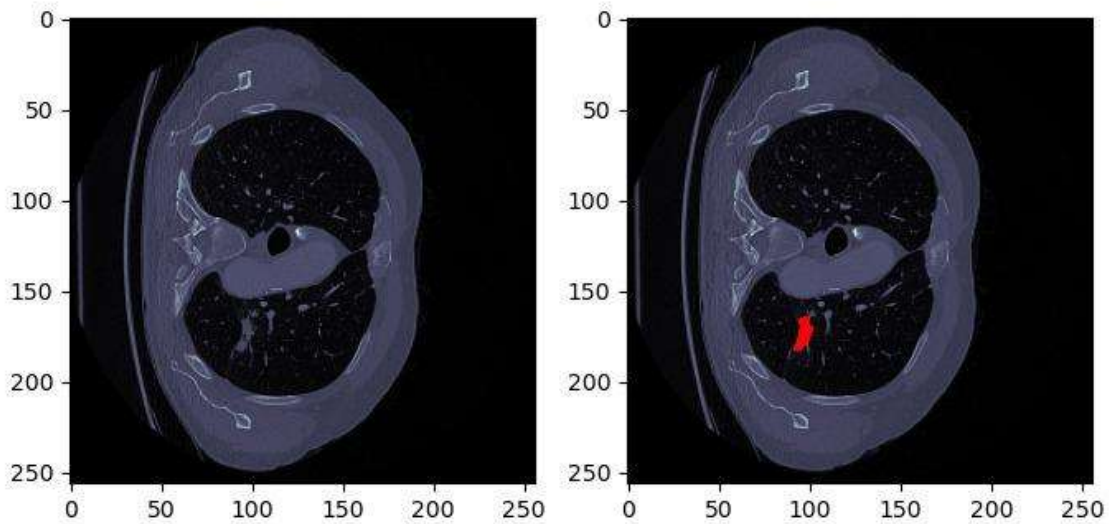
UNET Architecture

3.2 METHODS

The model.py file contains a UNet class, where we define our custom layers for this problem. In this particular case, we have used **3 layers for down-sampling**, **1 intermediate layer** and **3 layers of up-sampling** apart from the input and output layers, in total **7 layers**. We have coded each layer as a helper class in python, and this class contains a 2D convolution layer followed by ReLU and the same combination is used once again. Each layer is followed by a maxpool layer before connecting it to the next layer while downsampling and is concatenated after each layer of upsampling with its respective downsampling layer (to implement skip connection).

CHAPTER 4

RESULTS AND DISCUSSIONS



The above image highlights the presence of the tumor in the lungs. The tumor in the lung is segmented and shown explicitly using a mask.

UNet neural network was used to train the model in lung tumor segmentation for which it yielded a validation dice score of 0.7291593551635742. Validation dice score is the measure of how similar the objects are. A validation dice score of 72.9% is obtained when UNet is used to train the model to segment lung tumors from the dataset

The drive link below depicts the segmentation of the tumor present in the lungs in video format using the dataset images.

<https://drive.google.com/file/d/1Zb-3LON8XRCp12BjbY-QKDgKohI-CotK/view?usp=sharing>

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

Early detection of tumors in lungs can help the patient by enabling the use of non-invasive treatment options like medication and significantly increases the recovery rate. Use of neural network based segmentation techniques helps the doctors better diagnose the patients and lowers the chances of misdiagnosis. UNET models have shown significant increase in their accuracy and with the ongoing research, also promises higher degree accuracy in segmentation of medical imagery.

For the model developed in this project, an accuracy of 72.9% was achieved when segmenting lung MRI scans for detecting tumors.

For the future scope of this project, the model's accuracy would be further increased to detect tumors in their very early stages when they can often be missed by doctors during diagnosis.

CHAPTER 7

REFERENCES

- [1] GitHub - fshnkarimi/LungTumor-Segmentation: Automatically segment lung cancer in CTs
- [2] Dataset - Medical Segmentation Decathlon (medicaldecathlon.com)
- [3] W. Rahane, H. Dalvi, Y. Magar, A. Kalane and S. Jondhale, "Lung Cancer Detection Using Image Processing and Machine Learning HealthCare," 2018 International Conference on Current Trends towards Converging Technologies (ICCTCT), Coimbatore, India, 2018, pp. 1-5, doi: 10.1109/ICCTCT.2018.8551008.
- [4] R. P.R., R. A. S. Nair and V. G., "A Comparative Study of Lung Cancer Detection using Machine Learning Algorithms," 2019 IEEE International Conference on Electrical, Computer and Communication Technologies (ICECCT), Coimbatore, India, 2019, pp. 1-4, doi: 10.1109/ICECCT.2019.8869001.
- [5] R. Tekade and K. Rajeswari, "Lung Cancer Detection and Classification Using Deep Learning," 2018 Fourth International Conference on Computing Communication Control and Automation (ICCUBEA), Pune, India, 2018, pp. 1-5, doi: 10.1109/ICCUBEA.2018.8697352.
- [6] A. Rehman, M. Kashif, I. Abunadi and N. Ayesha, "Lung Cancer Detection and Classification from Chest CT Scans Using Machine Learning Techniques," 2021 1st International Conference on Artificial Intelligence and Data Analytics (CAIDA), Riyadh, Saudi Arabia, 2021, pp. 101-104, doi: 10.1109/CAIDA51941.2021.9425269.
- [7] S. Das and S. Majumder, "Lung Cancer Detection Using Deep Learning Network: A Comparative Analysis," 2020 Fifth International Conference on Research in Computational Intelligence and Communication Networks (ICRCICN), Bangalore, India, 2020, pp. 30-35, doi: 10.1109/ICRCICN50933.2020.9296197.
- [8] Q. Wu and W. Zhao, "Small-Cell Lung Cancer Detection Using a Supervised Machine Learning Algorithm," 2017 International Symposium on Computer Science and Intelligent Controls (ISCSIC), Budapest, Hungary, 2017, pp. 88-91, doi: 10.1109/ISCSIC.2017.22.
- [9] Wenqing Sun, Bin Zheng, and Wei Qian "Computer aided lung cancer diagnosis with deep learning algorithms", Proc. SPIE 9785, Medical Imaging 2016: Computer-Aided Diagnosis, 97850Z (24 March 2016); <https://doi.org/10.1117/12.2216307>
- [10] B. S, P. R and A. B, "Lung Cancer Detection using Machine Learning," 2022 International Conference on Applied Artificial Intelligence and Computing (ICAAIC), Salem, India, 2022, pp. 539-543, doi: 10.1109/ICAAIC53929.2022.9793061.